

HEALTHCONNECT CLINIC

Week 7 Analytics Testing & Refinement Report

Data Analytics Track

Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

AnalystLab Africa Experience Lab Internship Programme

Week 7

Final analytical report. Screenshot evidence is maintained separately in the companion Evidence Pack.

Executive Summary

Week 7 continued directly from the HealthConnect Week 6 advanced analytics dashboard. The focus moved from advanced development and validation into systematic testing, refinement and end-to-end validation. The existing dashboard and analytical outputs were tested against the underlying appointment data, including the four core dashboard KPIs, lead-time segmentation and interactions with previous no-show history, reminder channel, appointment type and distance to the clinic.

The validated baseline remained consistent: 5,000 appointments, 48.5% overall no-show rate, 60.5% 30+ day no-show rate, 55.4% repeat no-show rate and 53.7% lost slot rate. Lead-time analysis remained consistent, with observed no-show rates of 24.8%, 32.6%, 43.2% and 60.5% across the four lead-time bands.

Testing identified one substantive KPI functionality issue. The Repeat No-Show Rate initially remained at 55.4% when the Previous No-Show History slicer was set to “No prior no-show”. The DAX measure was refined so that the KPI responds to the slicer; re-testing returned 0.0% for the no-prior-no-show selection. Other observations were presentation or minor rounding matters rather than evidence that the underlying analysis was incorrect.

Dashboard refinement included removing the static Top 3 Insights section to create space for interactive analysis, rearranging the dashboard, adding interactive slicers, standardising user-facing labels and clarifying the second KPI as “30+ Day No-Show Rate”. The substantive Week 6 insights were retained in documentation.

Week 6 to Week 7 Transition

Week 6 produced the HealthConnect Advanced Analytics & Decision Support dashboard and extended the Week 5 descriptive analysis through deeper segmentation. The key Week 6 finding was the observed increase in no-show rates across booking lead-time bands, reaching 60.5% for appointments booked more than 30 days in advance. Within the 30+ day group, patients with previous no-shows recorded 67.9%, compared with 55.2% among patients with no previous no-show history.

Week 6 also examined Lead Time × Previous No-Show History, Lead Time × Reminder Channel, Lead Time × Appointment Type and Lead Time × Distance to Clinic. Week 7 therefore focused on whether the existing KPIs, calculations, filters, visual outputs and conclusions behaved as intended. The Data Science track was relevant because the combined long-lead-time and previous-no-show finding had been shared as a candidate interaction feature for predictive-model testing.

Week 7 Testing Objectives

- Validate important Week 6 KPI calculations against the underlying data.
- Check that dashboard values agree with validated calculations.
- Test relevant dashboard slicers and their effect on KPIs and visuals.
- Investigate unexpected or inconsistent analytical behaviour.
- Refine calculations or presentation where evidence identified a weakness.
- Re-test refined outputs.
- Reassess important findings across relevant segments.
- Document limitations and prepare Week 8 recommendations.

Data, Tools and Testing Basis

The same HealthConnect Appointment Dataset and Power BI analytical model were retained. Power Query was used for data preparation and calculated fields, while Power BI and DAX were used for KPI calculations and dashboard analysis.

Key fields included booking_lead_days, appointment_outcome, previous_no_shows, reminder_channel, appointment_type, distance_to_clinic_km and reminder_sent. Lead-time bands were standardised as 0–3 days, 4–14 days, 15–30 days and 30+ days. The 30+ category represents appointments booked more than 30 days in advance.

The dataset is fictional and synthetic. Findings are therefore intended for learning and analytical practice rather than direct clinical decision-making.

KPI Validation

KPI	Calculation	Dashboard	Validated Result	Status
Overall No-Show Rate	2,423 / 5,000 = 48.46%	48.5%	48.5%	PASS
30+ Day No-Show Rate	1,467 / 2,425 = 60.49%	60.5%	60.5%	PASS
Repeat No-Show Rate	1,152 / 2,079 = 55.41%	55.4%	55.4%	PASS
Lost Slot Rate	(2,423 + 263) / 5,000 = 53.72%	53.7%	53.7%	PASS

Dashboard Functionality Testing

Testing used controlled slicer selections and compared observed dashboard behaviour with validated values. The results below reflect the manual tests completed during Week 7.

Test	Scenario / Expected Result	Observed Result	Status / Action
Default / Clear State	All slicers = All; baseline 48.5%, 60.5%, 55.4%, 53.7%.	Baseline confirmed.	PASS — None
Lead Time = 30+	No-show 60.5%; Repeat 67.9%; Lost Slot 65.9%.	KPIs and visuals filtered correctly.	PASS — None
Previous No-Show = Has prior	No-show 55.4%; Repeat 55.4%; Lost Slot 59.6%.	Values and lead-time breakdown matched.	PASS — None
Reminder Channel	SMS, Email, None and WhatsApp selections tested.	All tested selections filtered consistently.	PASS — None
Appointment Type	Diagnostic, Follow-up, General and Specialist	All tested selections matched validated values.	PASS — None

	selections tested.		
Previous No-Show = No prior	No-show 43.5%; 30+ 55.2%; Repeat 0.0%; Lost Slot 49.5%.	Repeat initially stayed 55.4%; after DAX refinement it returned 0.0%.	PASS after refinement — DAX refined and re-tested
Lead Time selections	0–3 24.8%; 4–14 32.6%; 15–30 43.2%; 30+ 60.5%.	All four no-show rates matched; 0–3 lost-slot value re-checked and confirmed correct (31.45% rounds to 31.4%).	PASS — No calculation change
Distance to Clinic category order	Chart legend/category order expected to follow logical distance progression.	Original order was <5km, 15km+, 5-15km, Unknown; underlying values correct.	PASS after refinement — Custom sort applied in Power Query and re-tested

Lead-Time Functionality Results

Lead Time	No-Show	30+ KPI	Repeat	Lost Slot
0–3 days	24.8%	60.5%	35.0%	31.4%
4–14 days	32.6%	60.5%	37.0%	37.2%
15–30 days	43.2%	60.5%	49.6%	48.3%
30+ days	60.5%	60.5%	67.9%	65.9%

The 30+ Day No-Show Rate card is a segment-specific KPI. Its remaining at 60.5% when other lead-time bands are selected is not treated as an error because the card is explicitly defined around the 30+ segment.

Issues Identified and Solutions

Repeat No-Show KPI Slicer-Response Issue

When “No prior no-show” was selected, three KPIs responded correctly, but Repeat No-Shows remained at 55.4%. The original DAX measure explicitly filtered on the categorical field 'Previous No-Show History' = "Has prior no-show", causing the KPI calculation not to reflect the selected history context.

The DAX measure was refined to recognise the Previous No-Show History selection. “No prior no-show” now returns 0.0%, “Has prior no-show” retains the repeat-no-show calculation, and the All state retains the 55.4% baseline. Re-testing returned 0.0% for the no-prior-no-show selection, confirming successful resolution.

Lost Slot Rate Rounding Re-Check (0–3 Days)

For 0–3 days, the dashboard displayed 31.4%. The independent calculation from the raw dataset is 31.45%, which also rounds to 31.4% at one decimal place. On re-check, there was no rounding or display discrepancy; the two values agree. This line is retained to record that the check was performed.

Distance-to-Clinic Category Sort Order

The Lead Time × Distance to Clinic chart displayed categories in the order <5km, 15km+, 5-15km, Unknown, which does not follow a logical distance progression and made the chart harder to read correctly at a glance. This was identified as a presentation issue rather than a data or calculation error; the underlying values were independently validated and were correct. A custom sort order (<5km, 5-15km, 15km+, Unknown) was applied to the distance category field in Power Query so the chart now displays in logical distance order. The chart was re-tested after the fix and the category order and underlying values were confirmed correct.

Dashboard Refinements

- Removed the static Top 3 Insights section to create space for interactive analysis.

- Corrected the distance-to-clinic category sort order to a logical distance progression (<5km, 5-15km, 15km+, Unknown).
- Rearranged the existing Week 6 dashboard rather than rebuilding it as a new dashboard.
- Added interactive slicers for Lead Time, Previous No-Show History, Reminder Channel and Appointment Type.
- Standardised user-facing slicer labels.
- Standardised lead-time labels as 0–3 days, 4–14 days, 15–30 days and 30+ days.
- Clarified the second KPI as “30+ Day No-Show Rate”.
- Retained substantive Week 6 insights in written documentation after removing the static insight panel.
- Clarified percentage-point comparisons using “pp” where applicable.

Analytical Validation and Reassessment

- Overall no-show rate remained 48.5% across 5,000 appointments.
- No-show rates across lead-time bands remained 24.8%, 32.6%, 43.2% and 60.5%.
- Within 30+ days, previous no-show history was associated with 67.9% versus 55.2% for no previous no-show history.
- Within 30+ days, Follow-up appointments recorded an observed 65.1% no-show rate.
- Within 30+ days, the 15km+ group recorded an observed 66.4% no-show rate.
- Reminder-channel differences remained descriptive and were not interpreted as causal effects.

These are observed associations and segmentation signals, not causal conclusions. Distance was not treated as a simple monotonic relationship because the 5–15km group was lower than the <5km group in the 30+ segment.

Cross-Track Testing and Validation

The Week 6 Data Analytics finding concerning extended booking lead time combined with previous no-show history was shared with the Data Science track. Data Science confirmed that booking lead days and historical no-show rate were currently treated as independent signals and identified an interaction feature as a candidate for Week 7 testing.

The contribution can therefore be described as: Data Analytics identified the combined high-observed-risk segment → Data Science identified an interaction feature as a modelling opportunity → Week 7 provides a basis for testing whether that interaction adds predictive value.

The descriptive finding does not establish causation and does not claim that the interaction has already improved model performance.

Evidence Register

Screenshot evidence has been separated from this analytical report and is maintained in the companion document, “HealthConnect Clinic | Week 7 Testing & Refinement Evidence Pack”.

Evidence	Purpose	Companion Pack Section
E01	Final refined dashboard; dashboard rearrangement and slicers.	E01
E02	Default/all-filter dashboard state; baseline.	E02
E03	30+ day screenshot; lead-time filter functionality.	E03
E04	No prior no-show screenshot; corrected	E04

	Repeat No-Show KPI.	
E05	DAX before/after screenshot; issue and refinement evidence.	E05
E06	Slack/Data Science screenshot; cross-track validation evidence.	E06

Assumptions, Limitations and Remaining Issues

- The dataset is synthetic and intended for learning/project practice.
- Observed associations should not be interpreted as causal effects.
- Waiting-time values were previously identified as potentially unreliable because values exist for some no-show and cancelled appointments despite non-attendance.
- Distance to clinic and waiting-time fields contain some missing values.
- The 0–3 days Lost Slot Rate was re-checked against the raw dataset (31.45%, rounding to 31.4%) and confirmed to match the dashboard; no discrepancy or correction was required.
- The Repeat No-Show KPI required a DAX refinement and was successfully re-tested.
- The distance-to-clinic category sort order required correction in Power Query and was re-tested; underlying values were unaffected throughout.
- Further predictive testing of the Lead Time × Previous No-Show interaction belongs to the Data Science testing process.

Updated Business Insights

- Booking lead time remains the clearest observed segmentation pattern, increasing from 24.8% at 0–3 days to 60.5% at 30+ days.
- Previous no-show history provides additional segmentation value, particularly within the 30+ day group.
- The combined 30+ day and previous-no-show segment recorded the highest observed no-show rate in the core interaction analysis at 67.9%.
- Follow-up appointments and the 15km+ distance segment are additional observed segments for operational attention within the 30+ day group.
- Reminder-channel differences should be monitored descriptively and should not be interpreted as causal evidence.

Recommendations for Week 8

- Carry the validated four-KPI dashboard into final integration.
- Use the refined Repeat No-Show Rate measure in the final dashboard.
- Preserve the interactive slicer structure, clearer KPI title and corrected distance-to-clinic sort order.
- Use the validated Lead Time × Previous No-Show finding as an input for final cross-track integration.
- Evaluate the proposed interaction feature using appropriate Data Science model-performance metrics.
- Maintain clear distinction between descriptive associations and causal claims.
- Insert final evidence, update the repository and complete version-controlled documentation before Week 8 presentation.

Week 8 Readiness

The Data Analytics component has been systematically tested, the identified KPI interaction issue has been corrected and re-tested, and the main analytical findings remain consistent with the validated dataset. Remaining work is final evidence insertion, repository/documentation completion and final cross-track integration.

Conclusion

Week 7 moved the HealthConnect Data Analytics work from advanced analysis into systematic testing and refinement. KPI calculations and principal findings were validated, dashboard slicers were tested across relevant selections, and a genuine KPI interaction issue was identified and corrected. Dashboard usability was improved through interactive slicers, clearer labelling and removal of a static insight panel while retaining the analytical insights in documentation.

Appendix: Corrected Repeat No-Show Rate DAX

```
Repeat No-Show Rate =
VAR SelectedHistory =
    SELECTEDVALUE('Appointments'[Previous No-Show History])
RETURN
SWITCH(
    SelectedHistory,
    "No prior no-show", 0,
    "Has prior no-show",
        DIVIDE(
            CALCULATE([Total Appointments],
                'Appointments'[previous_no_shows] > 0,
                'Appointments'[appointment_outcome] = "No-Show"),
            CALCULATE([Total Appointments],
                'Appointments'[previous_no_shows] > 0), 0),
    DIVIDE(
        CALCULATE([Total Appointments],
            'Appointments'[previous_no_shows] > 0,
            'Appointments'[appointment_outcome] = "No-Show"),
        CALCULATE([Total Appointments],
            'Appointments'[previous_no_shows] > 0), 0)
)
```